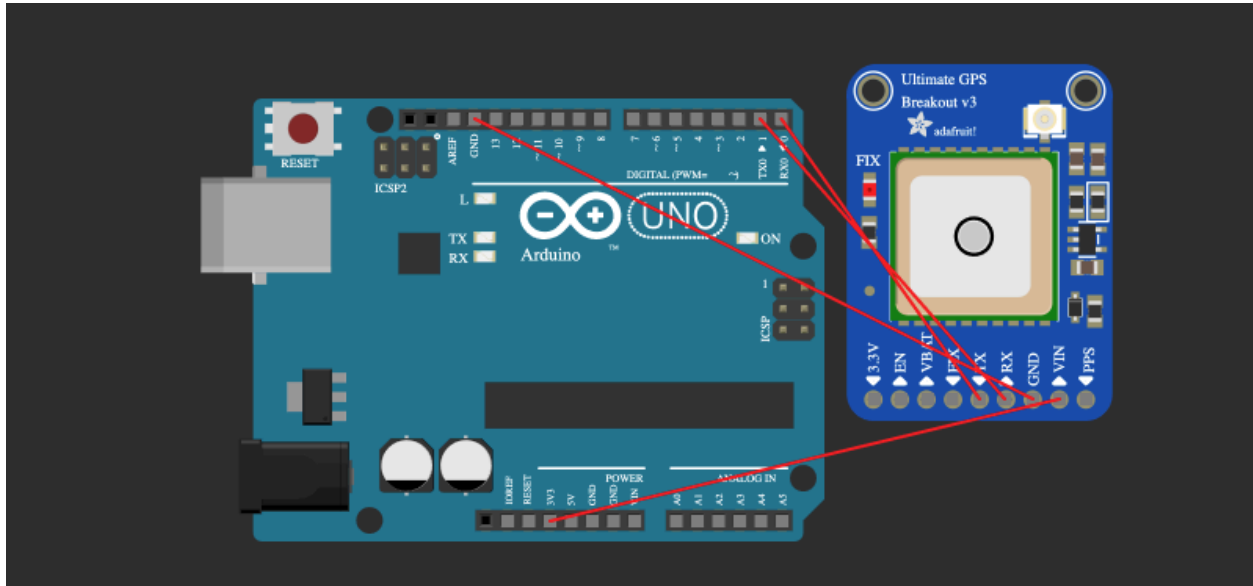


GPS:



DISCLAIMER: 3.3v is an output pin, VIN is the actual power input. To power GPS, we will need to connect the 3.3v pin on Arduino to VIN pin on GPS.

Components: 1 Arduino, 1 GPS

Arduino code:

```
#include "Adafruit_GPS.h"
Adafruit_GPS GPS;
void setup() {
  Serial.begin(9600);
}
void loop() {
  if (Serial.available() > 0) {
    string gpsResult = GPS.read();
    Serial.println(gpsResult);
  } else {
    Serial.println("No data");
  }
  delay(1000);
}
```

Networking:

Create two projects in Web Frontend: nw-sender and nw-receiver.

Project 1: nw-sender

Components: 1 Arduino

Arduino code:

```
void setup() {  
    pinMode(0, OUTPUT);  
    Serial.begin(9600);  
}  
void loop() {  
    string msg = "Your message here";  
    Serial.sendMessage(0, msg);  
    delay(2000);  
}
```

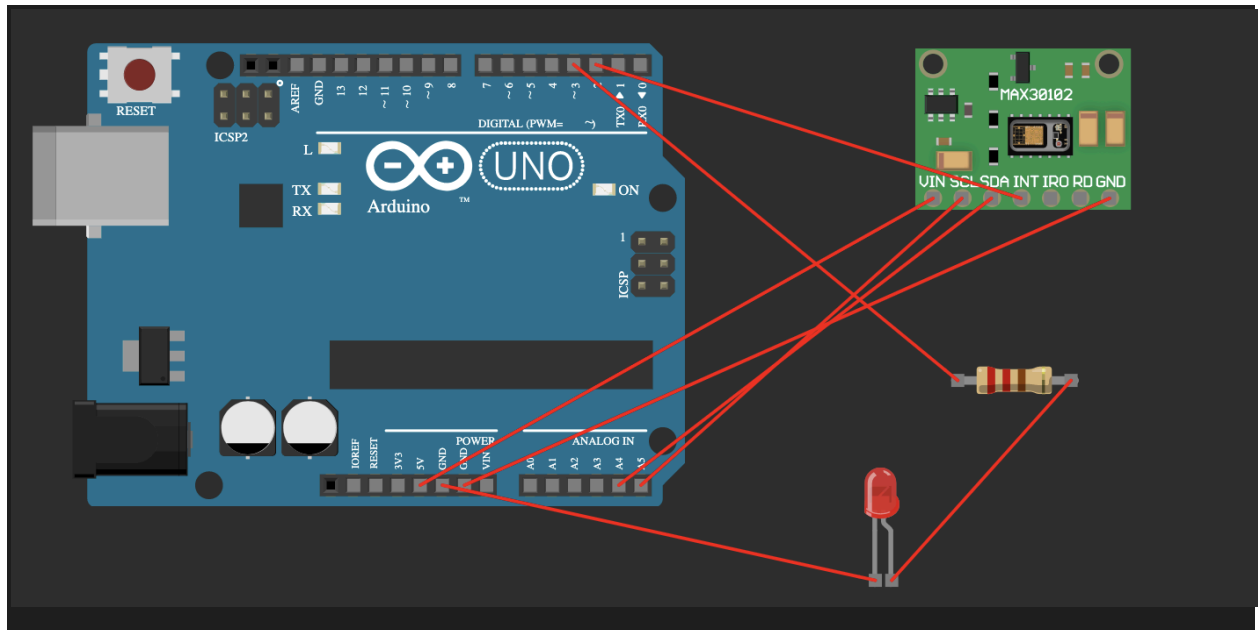
Project 2: nw-receiver

Components: 1 Arduino

Arduino code:

```
void setup() {  
    Serial.begin(9600);  
}  
void loop() {  
    if (Serial.availableMessage() != 0) {  
        string mystr = Serial.readMessage();  
        Serial.println("Printed by receiver: " + mystr);  
    }  
    delay(2000);  
}
```

PulseOximeter (Vitals):



Components: 1 Arduino, 1 Vitals Sensor

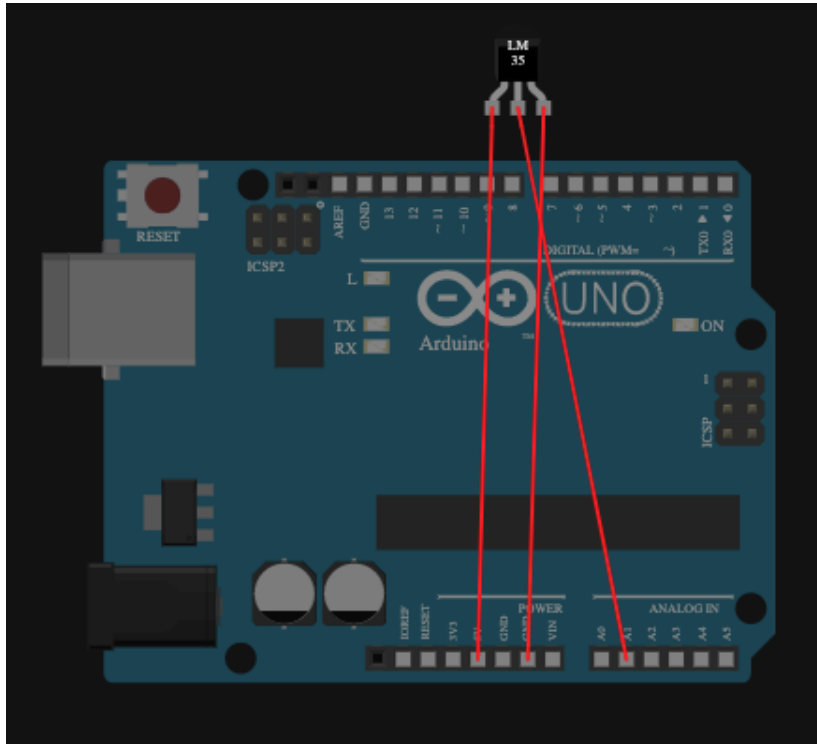
Arduino code:

```
#include "PulseOximeter.h"
PulseOximeter pox;

void setup() {
  pinMode(2, INPUT);
  pinMode(3, OUTPUT);
  Serial.begin(9600);
}

void loop() {
  if (pox.begin()) {
    string spO2_res = pox.getSpO2();
    string hb_res = pox.getHeartRate();
    Serial.println("Oxygen percentage: " + spO2_res + "; Heart rate: " + hb_res);
    digitalWrite(3, HIGH);
  } else {
    Serial.println("No data");
    digitalWrite(3, LOW);
  }
  delay(1000);
}
```

Temperature:

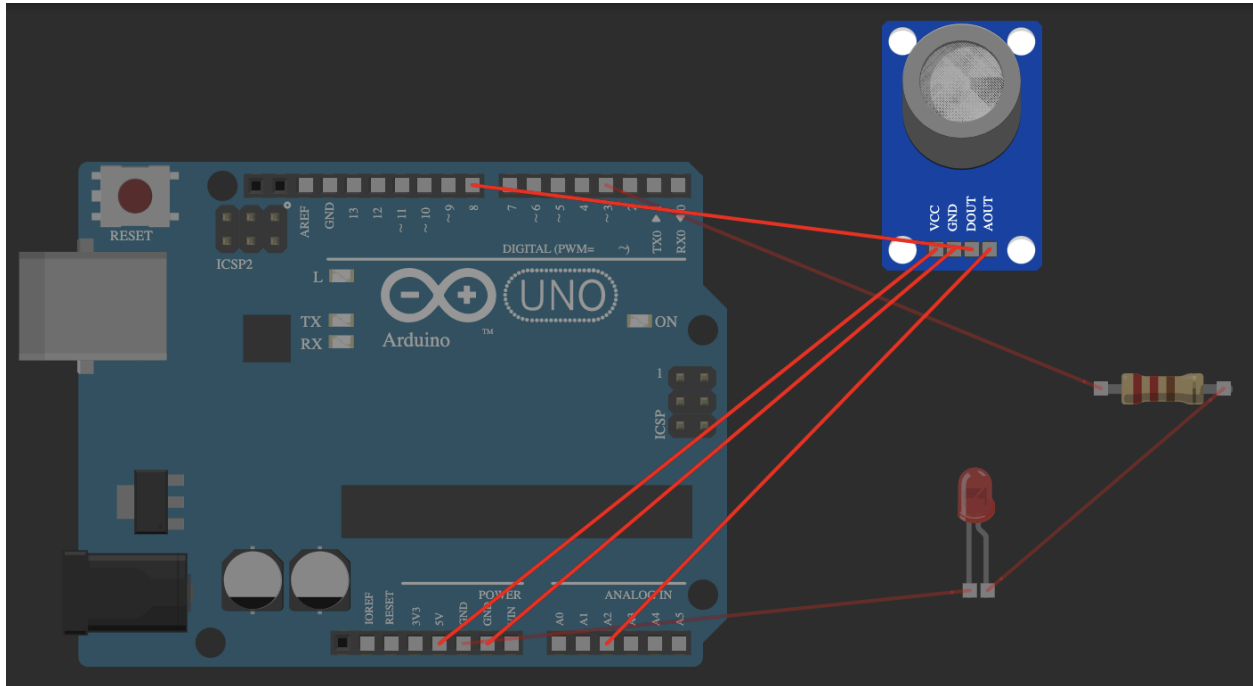


Components: 1 Arduino, 1 Temperature Sensor

Arduino Code:

```
void setup() {  
    pinMode(A1, INPUT);  
    Serial.begin(9600);  
}  
  
void loop() {  
    float temp = analogRead(A1) / 1023.0 * 5.0 * 100.0; // temperature = analog  
voltage * 100  
    Serial.println("temperature: " + to_string(temp));  
    delay(1000);  
}
```

Air Quality Sensor:



Components: 1 Arduino, 1 Air Quality Sensor, 1 LED, 1 resistor

Arduino Code:

```
#include "MQ135.h"

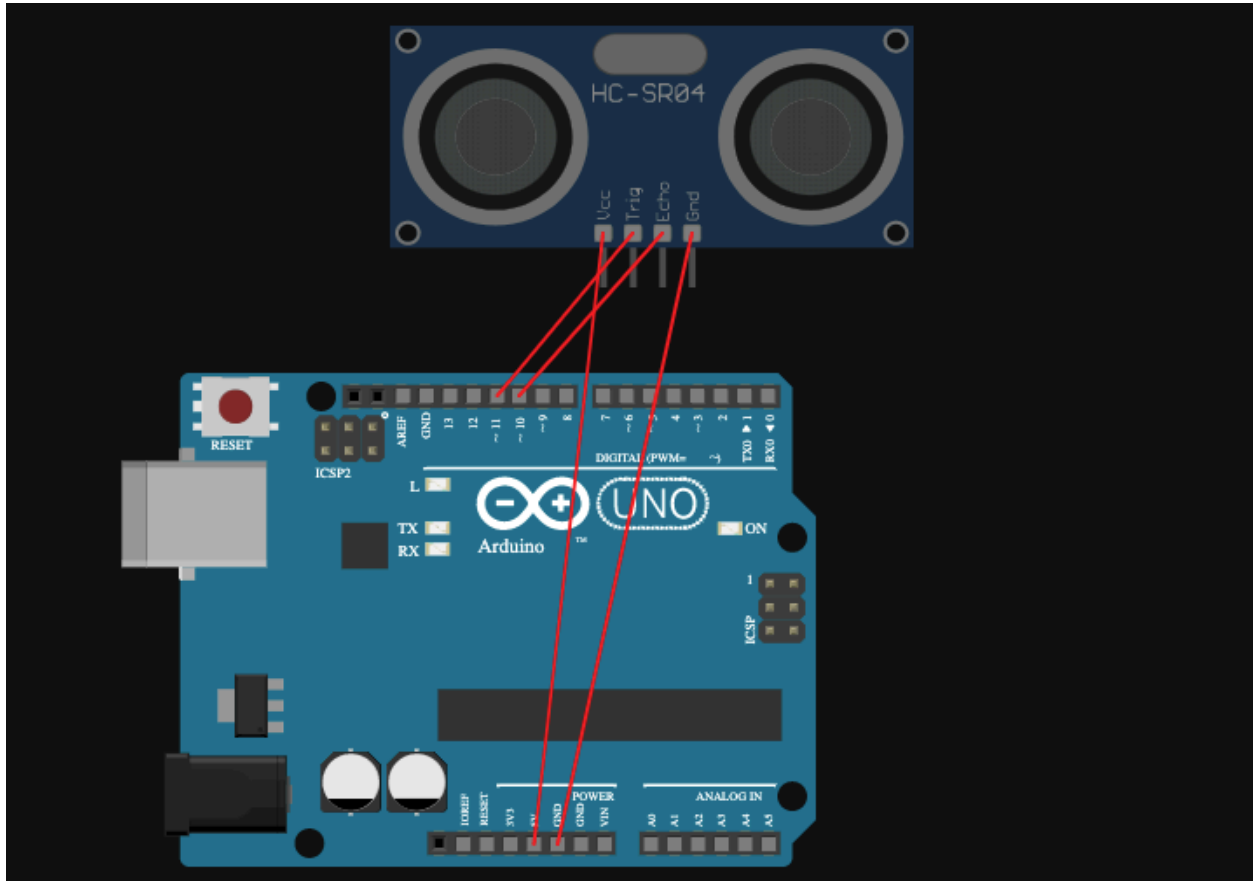
MQ135 mq135(A2);

void setup() {
  pinMode(A2, INPUT);
  pinMode(3, OUTPUT);
  pinMode(8, INPUT);
  Serial.begin(9600);
}

void loop() {
  float ppm = mq135.getPPM();
  Serial.println("ppm: " + to_string(ppm));

  int digitalResult = digitalRead(8);
  if (digitalResult == HIGH) digitalWrite(3, HIGH);
  else digitalWrite(3, LOW);
  delay(1000);
}
```

Ultrasonic Sensor:



Components: 1 Arduino, 1 Ultrasonic Sensor

Arduino Code:

```
const int trigPin = 11;
const int echoPin = 10;

float duration, dist;

void setup() {
  pinMode(trigPin, OUTPUT);
  pinMode(echoPin, INPUT);
  digitalWrite(trigPin, LOW); // Initialize trigPin as LOW
  Serial.begin(9600);
```

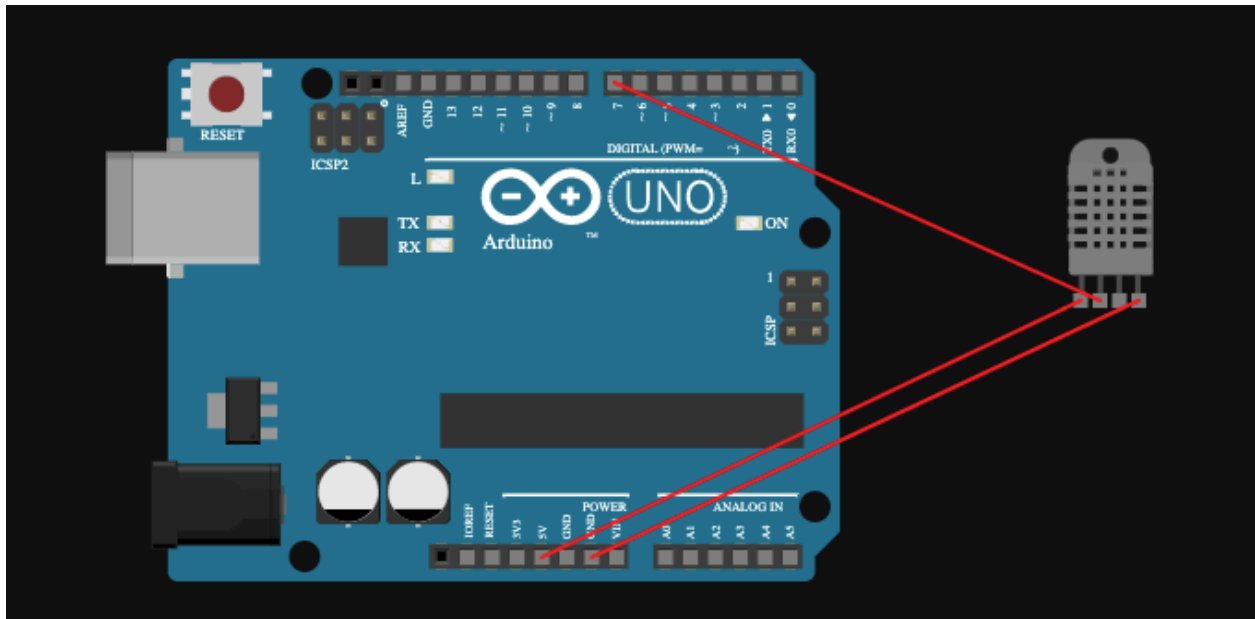
```

}

void loop() {
    digitalWrite(trigPin, HIGH);
    delayMicroseconds(10);
    digitalWrite(trigPin, LOW);
    duration = pulseIn(echoPin, HIGH); // µs
    dist = duration * pow(10, -6) * 34000 / 2; // cm
    Serial.println("Duration: " + to_string(duration) + "µs" + ", Distance: " +
to_string(dist) + "cm");
    delay(100);
}

```

Humidity Sensor:



Components: 1 Arduino, 1 Humidity Sensor

Arduino Code:

```

#include "DHT.h"
#define DHTPIN 7

DHT dht(DHTPIN);

void setup() {
    Serial.begin(9600);
    dht.begin();
}

```

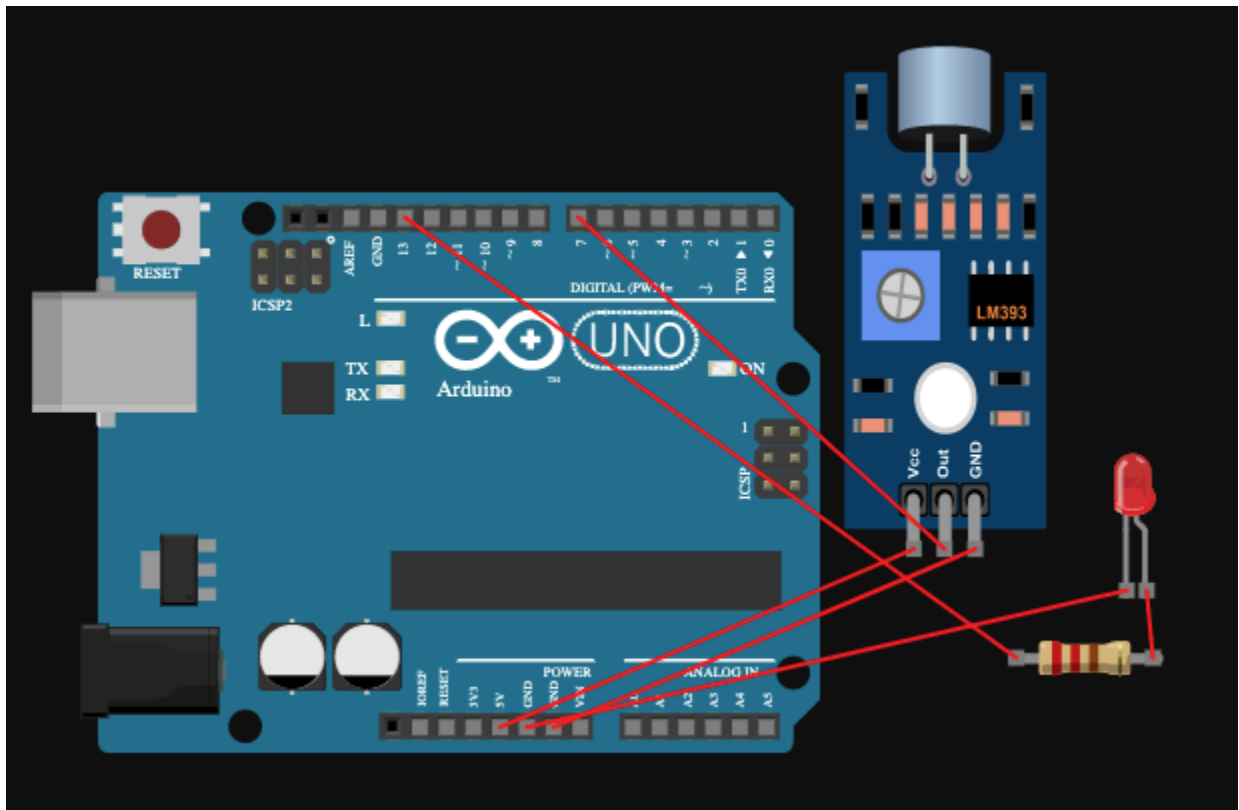
```

}

void loop() {
    bool isInCelcius = true;
    float h = dht.readHumidity() * 100; // Convert to %
    float t = dht.readTemperature(isInCelcius);
    Serial.print("Humidity: " + to_string(h) + "%; ");
    Serial.println("Temperature: " + to_string(t) + "°C");
    delay(1000);
}

```

Sound Sensor:



Components: 1 Arduino, 1 Sound Sensor

Arduino Code:

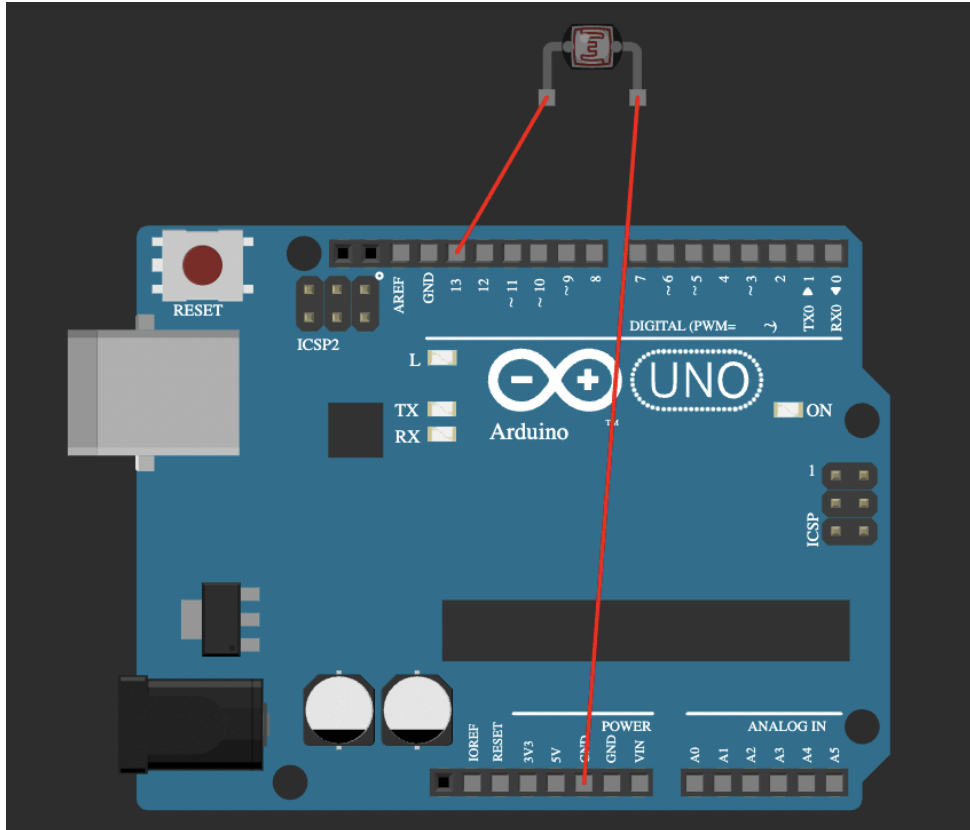
```
int ledPin=13;
```

```
int sensorPin=7;
int val = 0;

void setup() {
  pinMode(ledPin, OUTPUT);
  pinMode(sensorPin, INPUT);
  Serial.begin(9600);
}

void loop() {
  val = digitalRead(sensorPin);
  if (val==HIGH) {
    digitalWrite(ledPin, HIGH);
  } else {
    Serial.println(val);
    digitalWrite(ledPin, LOW);
  }
  delay(100);
}
```

LDR:



Components: 1 Arduino, 1 LDR

Arduino Code:

```
int ledPin=13;
float val = 0.0;

void setup() {
  pinMode(ledPin, OUTPUT);
  Serial.begin (9600);
}

void loop() {
  val = digitalRead(ledPin);
  Serial.println(val);
  delay(1000);
}
```

Functions Available:

- Libraries:

- DHT:

```
#include "DHT.h"
float readTemperature(bool S);
float readHumidity();
float convertCtoF(float c);
```

- PulseOximeter:

```
#include "PulseOximeter.h"
string getSpO2();
string getHeartRate();
int begin();
```

- Adafruit_GPS:

```
#include "Adafruit_GPS.h"
string read();
```

- MQ135:

```
#include "MQ135.h"
string read();
```

- Serial:

```
void begin(long speed);
int available();
int availableMessage();
int read();
string readString();
int write(int val);
int write(string str);
int sendMessage(int pin, string str, string device_addr);
int sendMessage(int pin, string str);
string readMessage();
int print(double val);
int print(string str);
int print(double val, int format);
int println(double val);
int println(double val, int format);
int println(string str);
```

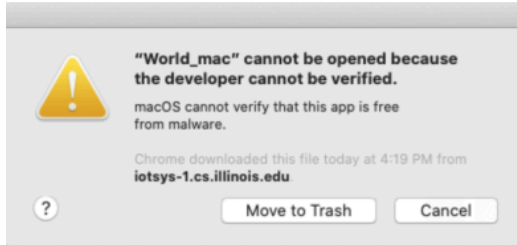
- Others:

```
void pinMode(uint8_t pin, uint8_t mode);
void digitalWrite(uint8_t pin, uint8_t value);
int digitalRead(uint8_t pin);
int analogRead(uint8_t pin);
void analogWrite(uint8_t pin, float value);
void delay(unsigned long ms);
void delayMicroseconds(unsigned int us);
double micros();
double millis();
void initializeTime();
unsigned long pulseIn(uint8_t pin, uint8_t value);
```

```
unsigned long pulseIn(uint8_t pin, uint8_t value, unsigned long timeout);
bool isAlpha(char thisChar);
bool isAlphaNumeric(char thisChar);
bool isAscii(char thisChar);
bool isControl (char thisChar);
bool isDigit(char thisChar);
bool isGraph(char thisChar);
bool isHexadecimalDigit(char thisChar);
bool isLowerCase(char thisChar);
bool isPrintable(char thisChar);
bool isPunct(char thisChar);
bool isSpace(char thisChar);
bool isUpperCase(char thisChar);
bool isWhitespace(char thisChar);
long random(long max);
long random(long min, long max);
```

Common Issues (please feel free to add more that you see!):

1. For Mac version's virtual world, if you see:



Open terminal and run:

```
chmod -R +x World_mac.app/<directory>
```

on the directory that you downloaded the executable to. Then open Application, go to "Security and Privacy" in Settings. Click on "Open Anyway".

2. Do not use the yellow LED!

I was going to use different leds for different sensors, but the components are different. You will get an error about components being incorrect.